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FGP-2000-AQ-1 Fractal Graphene Paste™ Technical Datasheet

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FGP-2000-AQ-1	Product: Fractal Graphene Paste™
Product Description: An aqueous dispersion of few-layer turbostratic graphene aggregates produced from carbon-rich gas explosion synthesis	
Benefits: • Pre-dispersed and stabilized graphene, remains evenly distributed, preventing agglomeration • Improves mechanical properties such as strength, durability, and functional characteristics of the end product • Uniform dispersion and controlled purity deliver reproducible results, reducing variability in production • Due to fine particle size, it integrates uniformly throughout the host material for optimal reinforcement	

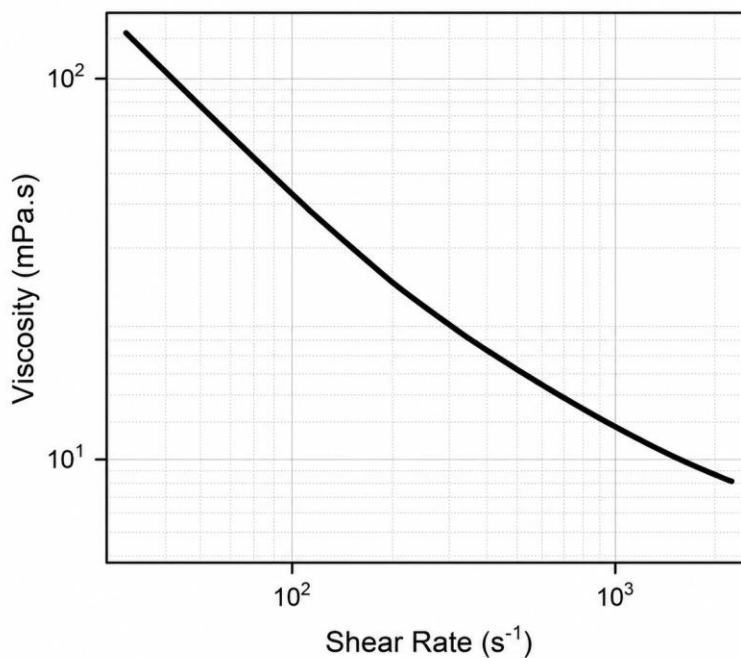
Product Information	
Raw Material	Fractal Graphene Aggregate
Forms of Materials	Liquid dispersion
Graphene Type	FGA-1
Functional Groups	None
SSA (BET)	130-180 m ² /g
Lateral Size (primary, TEM)	20-50 nm
Thickness	3-9 layers
Aspect Ratio	15:1



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Dispersion Properties	
Appearance	Flowable black liquid
Solvent	Deionized water
Dispersant	Anionic
Zeta Potential	-45 mV
S.G.	1.11 @ 20 wt.% FGA-1
Graphene Solids	20 wt.% FGA-1
Total Solids	24 wt.%
pH	8-9
Particle Size	D90 <500 nm, D50 =35 nm, D10 = 15 nm
Viscosity at 25°C	0.051 Pa.s at 100 s ⁻¹ 0.014 Pa.s at 1000 s ⁻¹
Compatibility	Water-based systems, such as concrete, aqueous coatings & polymer emulsions/dispersions.
Shelf life/storage	24 months from date of manufacture when stored unopened at 10-30°C,

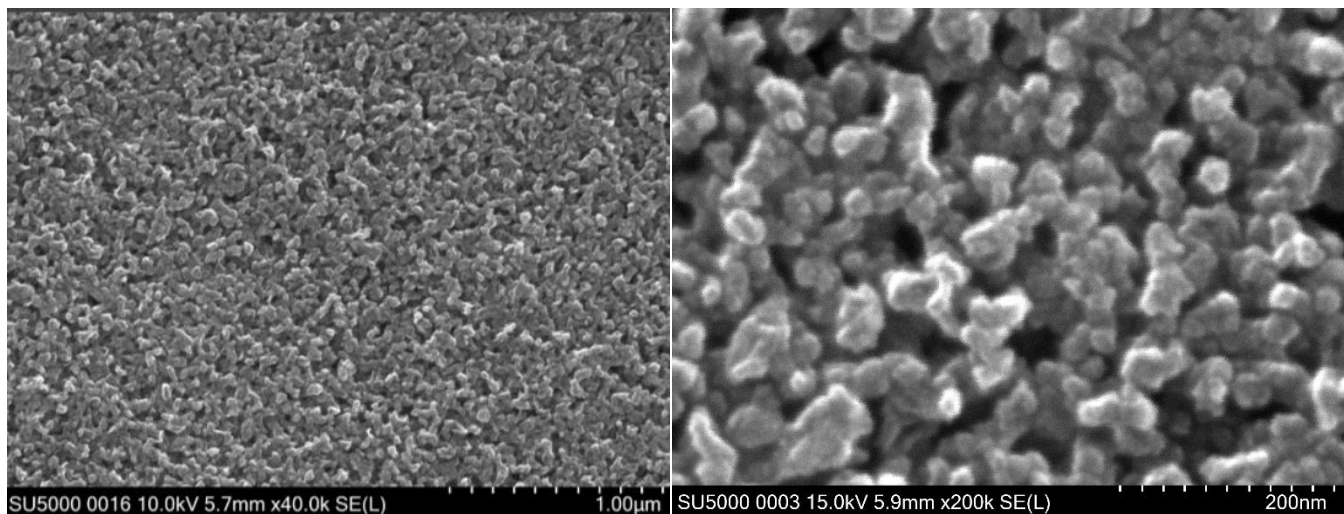
Rheology Curve





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Scanning Electron Micrographs



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